# Deep Spatiotemporal Subpixel Mapping Network by Integrating a Prior Fine Land Cover Map With Change Detection

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***Abstract*— Subpixel mapping (SPM) is a prevailing technique for addressing the mixed pixel issue by estimating the subpixel-scale distribution of land cover types within mixed pixels. Recently, spatiotemporal SPM (STSPM) has advanced traditional SPM by integrating prior fine spatial resolution (FR) images to alleviate the uncertainty in SPM results. However, most existing STSPM methods are typically model-driven or meticulously handcrafted, and they usually struggle to flexibly characterize the diverse and complex patterns of land cover patches. This article proposes a novel deep STSPM network (STSPMNet) that integrates a prior FR land cover map with bitemporal coarse spatial resolution (CR) remote sensing images using deep learning and change detection techniques. STSPMNet first combines super-resolution and semantic segmentation to achieve the end-to-end SPM from CR remote sensing images to FR land cover maps. A change-detection-based optimization approach is then developed to incorporate a prior FR land cover map with the estimated FR land cover probabilities to generate the final FR land cover map at the prediction date. Three experiments were conducted to evaluate the effectiveness of the proposed STSPMNet. Experimental results demonstrate that STSPMNet outperforms two SPM methods and two STSPM method by recovering more spatial details of land cover patches and generating higher accuracy metrics. Hence, STSPMNet offers a robust solution for producing FR land cover maps from CR remote sensing imagery.**

***Index Terms*— Change detection, generative adversarial networks (GANs), land cover, prior fine land cover map, semantic segmentation, subpixel mapping (SPM).**

## I. Introduction

LAND cover data are used to represent the occupation of Earth's surface by natural features and human activities, and they serve as essential inputs for a wide range of geoscience applications [1], [2], [3]. Extracting land cover information from remote sensing imagery is an efficient and widely adopted method that has generated numerous land cover products [4], [5], [6], [7]. Due to sensor limitations, mixed pixels containing multiple land cover types are ubiquitous in various remote sensing images, especially in low-resolution and medium-resolution images [8], [9], [10]. For instance, the diameter of trees is typically less than 10-m, and the width of roads is usually less than 30-m. Consequently, mixed pixels with composite spectral responses from diverse land cover types result in challenges for pixel-based classification [11], [12], [13]. Traditional hard pixel-based classifications assign each pixel with a single land cover type, resulting in inaccurate representations of mixed pixels [14], [15], [16]. To overcome the shortages of hard classification, soft classification (also known as spectral unmixing) produces fraction images that characterize the proportions of land cover types within a mixed pixel [17], [18]. Even though, it still cannot specify the spatial distribution of different land cover types within a mixed pixel [8], [19], [20], [21].

Subpixel mapping (SPM), also referred to as super-resolution mapping, was proposed as a prevailing way to addressing the mixed pixel problem in both traditional hard and soft classifications [21]. SPM initially divides each pixel into subpixels and then assigns land cover types to these subpixels within a mixed pixel under the constraint of maintaining area proportion for each land cover type [20], [21], [22]. This technique produces a hard-classified map with a higher resolution than the original input remote sensing images [23], [24], [25]. Over the past decades, numerous SPM methods have been developed, which can be categorized into four categories: 1) spatial dependence-based SPM methods; 2) geostatistical prior-based SPM methods; 3) spatial regularization-based SPM methods; and 4) auxiliary data-based SPM methods [11]. The first three categories primarily rely on coarse spatial resolution (CR) remote sensing imagery to extract spatial knowledge for determining the type of each subpixel within a mixed pixel [1], [21], [26], [27]. SPM is an intrinsically ill-posed problem due to the potential existence of multiple plausible configurations of land cover types within a mixed pixel when relying solely on spatial knowledge [11], [25], [28]. The last category aims to take full use of auxiliary data to reduce the uncertainty of SPM results [11], [22]. Various auxiliary data have been used in this category, including panchromatic images [29], [30], light detection and ranging data [31], multiple shifted images [32], [33], and prior fine spatial resolution (FR) land

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cover maps [28], [34]. To ensure land cover information consistency between the input CR remote sensing imagery and auxiliary data, the auxiliary data often need to be acquired at the same or around dates as the CR remote sensing imagery. This requirement for acquisition dates restricts the applicability of auxiliary data, except for prior FR land cover maps, which can be obtained either before or after the acquisition date of the CR remote sensing images; therefore, prior FR land cover maps are widely used as auxiliary data sources in SPM.

Ling et al. [28] introduced spatiotemporal SPM (STSPM), which extends traditional SPM by leveraging multitemporal images to enhance SPM results. STSPM usually incorporates a prior FR land cover map and its corresponding CR remote sensing image into a CR remote sensing image at the prediction date, generating land cover data at a subpixel scale that matches the prior FR land cover map. There are two main categories of STSPM methods: 1) spatiotemporal dependence-based STSPM methods and 2) change detection-based STSPM methods [28], [35]. The former category aims to explore dependence knowledge between and within pixels in both space and time dimensions [35]. In contrast, the latter category employs change detection techniques to compare bitemporal CR fraction images to estimate the land cover types of changed subpixels [28]. However, these STSPM methods are typically model-driven or meticulously handcrafted, rendering them too rigid to accurately characterize the diverse and complex patterns of land cover [28], [35], [36].

Recently, with the rapid advancements of deep learning, several deep learning-based SPM methods have been proposed as data-driven approaches to flexibly learn the complex patterns of land cover from exemplar pairs by leveraging the end-to-end mapping and feature representation capabilities of networks [37], [38]. A range of advanced deep learning models have been applied in SPM, including super-resolution reconstruction networks [39], [40], [41], semantic segmentation networks [42], [43], graph convolutional networks [44], attention mechanism-based networks [36], [45], and multiscale pyramid networks [46]. Although these deep learning-based SPM methods have achieved satisfactory performance, they still face two significant challenges. First, these methods may result in the smooth contours of land cover patches or even the loss of small-sized patches [47]. This issue arises because networks used in SPM mainly depend on stacked convolutional layers to expand the receptive field and learn deep-level features, resulting in the gradual blurring or even disappearance of small-sized objects (i.e., spatial details) from shallow-level to deep-level features [37]. Second, most deep learning-based SPM methods are dedicated to benefiting from deep learning networks but ignore the potential of auxiliary data to mitigate the uncertainty of SPM outcomes. These methods primarily rely on exemplar pairs of CR remote sensing or fraction images and their corresponding FR land cover labeled images to learn complicated nonlinear relationships [39], [40], [41]. However, the auxiliary data (e.g., prior FR land cover maps) commonly employed in conventional model-driven SPM methods are seldom integrated into deep learning-based SPM methods to improve the accurate representation of land cover within mixed pixels.

To address the inaccurate representation of land cover patches and the ignorance of prior FR land cover maps in existing deep learning-based SPM methods, this study develops a novel deep STSPM network (STSPMNet) that integrates a prior FR land cover map with bitemporal CR remote sensing images to generate a FR land cover map at the prediction date. Compared with the existing SPM and STSPM methods, STSPMNet has several features and contributions.

1) It extends conventional model-driven STSPM to data-driven STSPM by leveraging deep learning networks.
2) It combines super-resolution with semantic segmentation to achieve end-to-end SPM from CR remote sensing imagery to a FR land cover map.
3) Change-detection-based optimization approach is developed to integrate a prior FR land cover map acquired either before or after the prediction date into STSPMNet to reduce the uncertainty of SPM results.

The remainder of this article is organized as follows. Section II describes the proposed method. Section III illustrates the experimental results, and is discussed in Section IV. Section V presents conclusions.

## II. Methodology

The proposed STSPMNet is designed to obtain a FR land cover map $X$ from a CR remote sensing image $Y$ at the prediction date with the help of a CR-FR pair of images at the other date (i.e., before or after the prediction date). Similar to typical SPM, STSPMNet first divides each pixel in CR image $Y$ into $S \times S$ FR pixels (i.e., subpixels), and then it identifies the land cover type of each pixel in FR image $X$ while adhering to the fraction constraints. Assuming that the CR image $Y$ at the prediction date contains $I \times J$ pixels, the corresponding FR land cover map $X$ with $C$ types comprises $I \times S \times J \times S$ pixels using the scale factor $S$. Let $X_{\text{other}}$ be the prior FR land cover map with $C$ types derived from the FR remote sensing image $H_{\text{other}}$, which is acquired at the other date but the same region as $X$. Similarly, let $Y_{\text{other}}$ be the CR remote sensing image acquired at the same date as $X_{\text{other}}$. The CR and FR images are assumed to be spatially aligned.

With a CR remote sensing image at the prediction date and a CR-FR pair of images at the other date as inputs, STSPMNet operates through the following five steps, as shown in Fig. 1.

1) Implementing radiation normalization for input remote sensing images. Radiometric differences between remote sensing images may significantly affect the comparison of bitemporal CR remote sensing images and the super-resolution of CR images, and thus relative radiation normalization is necessary for both the bitemporal CR remote sensing images and the FR remote sensing image at the other date.
2) Designing STSPMNet architecture. To obtain end-to-end SPM results, two processes of super-resolution and semantic segmentation are combined into STSPMNet.
3) Training STSPMNet. The CR remote sensing image at the other date, along with the prior FR land cover map

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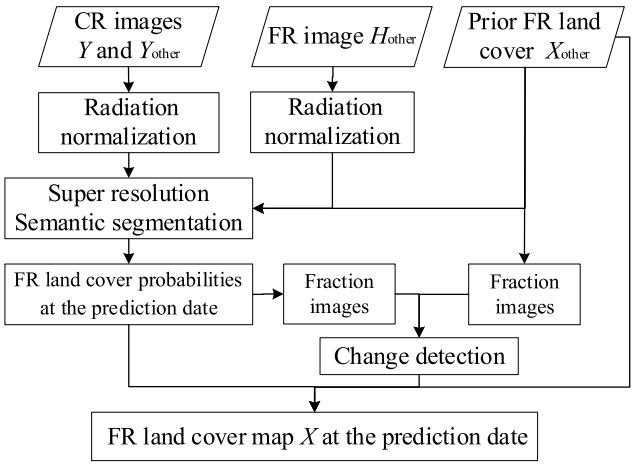



Fig. 1. Flow chart of STSPMNet.

and its corresponding FR remote sensing images are employed to train STSPMNet.

4) Predicting FR land cover map at the prediction date. With the trained STSPMNet, the CR remote sensing image at the prediction date is used as input to estimate the probability of land cover types for each FR pixel.
5) Incorporating the prior FR land cover map into the estimated FR land cover probabilities to obtain the final FR land cover map at the prediction date.

Detailed implementation is provided below.

### A. Radiation Normalization

Remote sensing images captured by different satellite sensors or at different dates can lead to radiometric differences that impact the subsequent analysis [48]. A common strategy to eliminate these radiometric differences is to use relative radiometric normalization, which establishes a mapping relationship between different resolution remote sensing images using techniques such as regression-based algorithms and histogram matching [49]. The typical histogram matching is exploited here to eliminate the radiometric differences between input remote sensing images.

### B. STSPMNet Architecture

STSPMNet is developed to provide end-to-end SPM results, and its architecture is illustrated in Fig. 2. The primary objective of STSPMNet is to enhance the spatial resolution of input CR remote sensing images and produce FR classified results. To achieve this, both super-resolution and semantic segmentation are essential for STSPMNet. Super-resolution is performed using a typical super-resolution generative adversarial network (SRGAN), which enhances the spatial resolution of the input CR images to generate FR remote sensing images [50]. Subsequently, semantic segmentation is applied to the super-resolved images to generate classified results. The multiscale UNet (MUNet) is designed as the semantic segmentation component within STSPMNet. Although the typical UNet is lightweight and effective, it cannot adequately capture features for different sizes of land cover patches [51]. To address this, the atrous spatial pyramid pooling (ASPP), inspired by the DeepLabV3 model [52], is integrated into UNet to improve the performance of semantic segmentation.

*1) Super-Resolution Generative Adversarial Network (SRGAN):* The super-resolution component of STSPMNet is based on generative adversarial networks (GANs). A GAN typically contains a generator and a discriminator, which engage in a zero-sum game where they compete against each other to improve their performance [50], [53]. STSPMNet also has a generator $G$ and a discriminator $D$ to operate adversarial learning and optimize their performance through a function Optimal($G$, $D$). SRGAN can be mathematically expressed by

$$\text{Optimal}(G, D) = \min_G \min_D L_{\text{SRGAN}} \tag{1}$$

$$L_{\text{SRGAN}} = L_G + L_D \tag{2}$$

where $G$ is the generator, $D$ is the discriminator, $L_{\text{SRGAN}}$ is the sum of the generator loss function $L_G$ and the discriminator loss function $L_D$. In the learning, $G$ and $D$ obtain the ability to generate realistic data and distinguish the predicted data from the truth data, respectively. SRGAN model expects that $G$ will be able to generate data that cannot be discriminated by $D$.

The proposed STSPMNet employs SRGAN to generate FR remote sensing images from CR remote sensing images. The generator network of SRGAN aims to produce super-resolved remote sensing images. The discriminator networks of SRGAN receive either a fake or a real FR image as the input to distinguish its source.

*a) Generator network:* The generator network is responsible for feature learning and super-resolution reconstruction, as shown in Fig. 2. This network involves 16 residual modules that are designed to extract features, while subpixel convolution layers are leveraged to increase the spatial resolution of the input CR remote sensing images [50]. To improve information flow across layers and reduce the vanishing gradient issue that can arise with deeper networks, skip connections are utilized between the residual modules. Unlike conventional interpolation-based upsampling for increasing resolution, the PixelShuffle layer (i.e., subpixel convolution layer) is utilized to achieve the desired increase in resolution by applying a specific upscaling factor [50]. The mathematical expression for mapping the CR image $Y$ to the FR image $H$ is provided as

$$H = G(Y) \tag{3}$$

where $G()$ represents the generator network of SRGAN.

*b) Discriminator network:* The discriminator network is essential in the adversarial training process, as it differentiates between the super-resolved images derived from the generator and the actual FR images. The standard discriminator in GANs is used in SRGAN [50], [54]. It is similarly adopted as the discriminator of STSPMNet by

$$D(H) = \sigma(C(H)) \to 1 \quad \text{Real data?} \tag{4}$$

$$D(H_{\text{SR}}) = \sigma(C(H_{\text{SR}})) \to 0 \quad \text{Fake data?} \tag{5}$$

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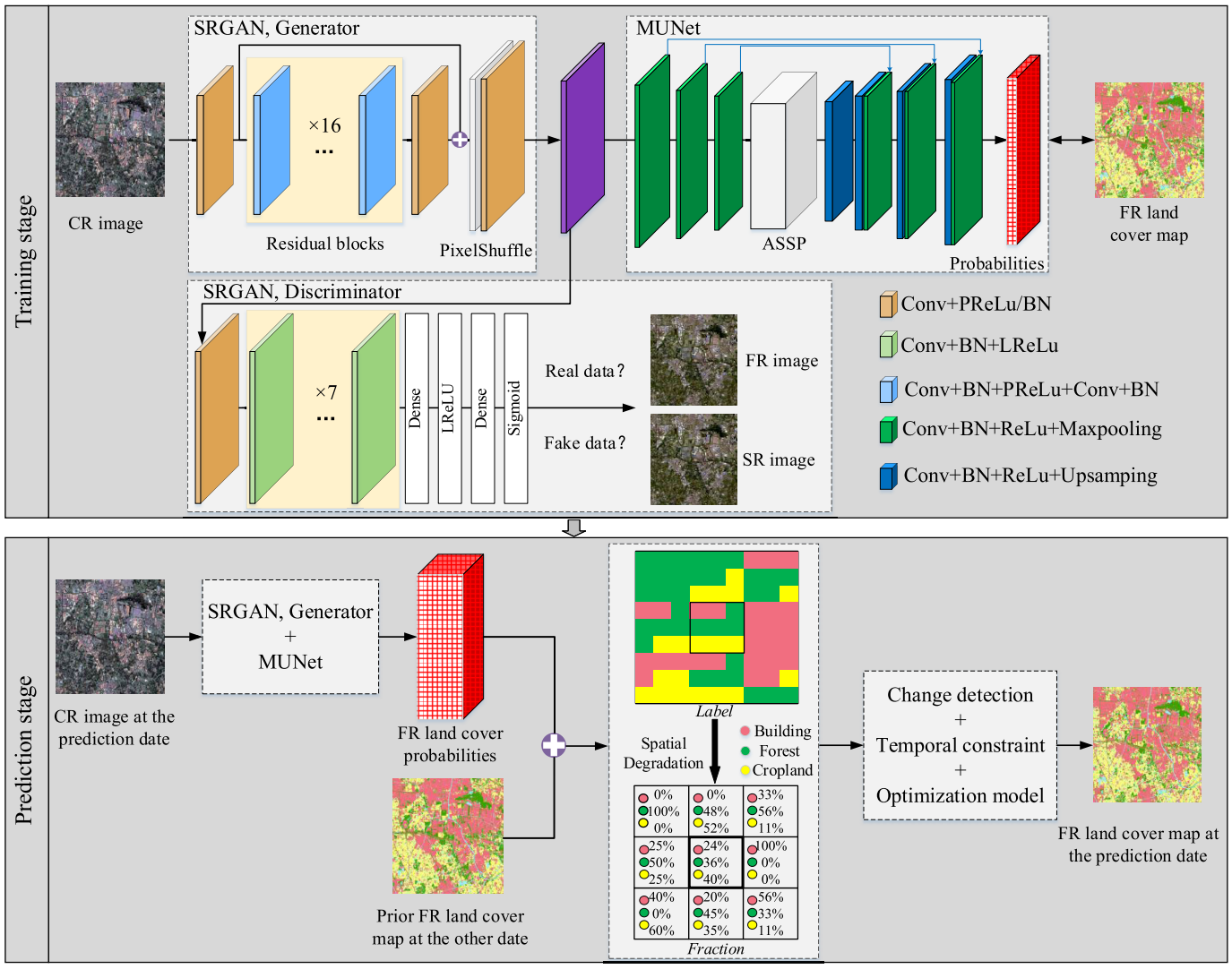



Fig. 2. STSPMNet architecture. SRGAN is used to increase the spatial resolution of CR image at the other date and MUNet generates FR land cover maps during training stage. In prediction stage, the trained SRGAN and MUNet are first employed to predict the FR land cover map at the prediction date and then it is combined with the prior FR land cover map by change detection and optimization model.

where $D()$ is the discriminator network of SRGAN, $H$ is the real FR remote sensing image, $H_{\mathrm{SR}}$ is the super-resolved image by the generator with the input of CR remote sensing image, $\sigma()$ and $C()$ represent the sigmoid function and the nontransformed output of discriminator, respectively.

*2) Multiscale UNet (MUNet):* MUNet modifies and extends the traditional UNet architecture by integrating an ASPP module, and it is used as the semantic segmentation component of STSPMNet, as depicted in Fig. 2. MUNet retains the U-shaped structure and includes a contracting path to characterize contextual information and a symmetric expanding path that facilitates precise localization. The central part of UNet architecture is replaced by ASPP, which enhances the ability of MUNet to extract multiscale features by effectively capturing different sizes of land cover patches. ASPP adopts multiple atrous convolutional layers with varying dilation rates but identical filter parameters to modify the field-of-view of filters for capturing features across multiple scales [52], [55], as shown in Fig. 2. MUNet takes the output of SRGAN to predict FR land cover probabilities.

*3) Loss Functions:* The loss function is critical for quantifying the error or discrepancy between the model prediction and the ground truth. To train the proposed STSPMNet, we combine the loss functions from SRGAN with the widely used cross-entropy loss for MUNET to form the total loss function of STSPMNet.

The SRGAN includes content loss, perceptual loss, and adversarial loss [50], [53], [56]. Content loss aims to preserve pixel-level accuracy [50]. Content loss measured by mean squared error (mse) is utilized to maximize the similarity between the super-resolved image $H_{\mathrm{SR}}$ and the real FR image $H$ during the training of SRGAN. The content loss function with mse is formulated as

$$L_{\mathrm{MSE}} = \frac{\sum_{b=1}^{B}\sum_{i=1}^{S\times I}\sum_{j=1}^{S\times J}\left(H_{\mathrm{SR},ijb} - H_{ijb}\right)^2}{B \times S^2 I J} \tag{6}$$

where $B$ represents the channel number in the FR image.

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Content loss based on mse computed in pixels alone is limited in capturing high-frequency details in images. Therefore, the perceptual loss function was introduced to enhance semantic similarity and the capture of high-frequency details and improve feature learning, thereby providing a more accurate perception of real images [50]. Most current super-resolution models measure the perceptual loss by a convolutional layer before the fifth max-pooling layer of a pretrained VGG19 network [50], [53], [56], aiming to capture perceptual differences at higher feature levels [57]. The super-resolved image $H_{\mathrm{SR}}$ and the real FR image $H$ are fed into the $k$th convolutional layer of the pretrained VGG19 network to calculate the derivation of perceptual loss. The perceptual loss function is formulated as

$$L_{\mathrm{Per}} = \frac{\sum_{b=1}^{B_k}\sum_{i=1}^{h_k}\sum_{j=1}^{w_k}\left(\mathrm{VGG19}(H)_{ijb} - \mathrm{VGG19}(H_{\mathrm{SR}})_{ijb}\right)}{h_k \times w_k} \tag{7}$$

where $\mathrm{VGG19}(H)_{ijb}$ and $\mathrm{VGG19}(H_{\mathrm{SR}})_{ijb}$ separately denote the feature outputs at position $(i, j)$ for channel $b$ of $H_{\mathrm{SR}}$ and $H$ from the VGG19 network, $h_k, w_k$, and $B_k$ represent the height, width, and number of channels for the features at the $k$th layer, respectively. The optimal layer is chosen by sensitivity analysis in Section IV-A.

Adversarial loss is often used in GANs to help generator and discriminator networks to generate realistic textures through discriminator-guided training, leading to the generation of high-quality data by the generator. The derivation of adversarial loss depends on the discriminator network of SRGAN. The adversarial loss function of the generator $L_{\mathrm{adv}}^{G}$ is calculated by

$$L_{\mathrm{adv}}^{G} = -\log(D(H_{\mathrm{SR}})) \tag{8}$$

where $D(H_{\mathrm{SR}})$ is the probability of discriminator for the super-resolved image $H_{\mathrm{SR}}$.

The adversarial loss function of the discriminator $L_{\mathrm{adv}}^{D}$ is formulated as

$$L_{\mathrm{adv}}^{D} = \log(D(H) - 1) + \log(D(H_{\mathrm{SR}})) \tag{9}$$

where $D(H)$ is the output probability of discriminator for the real image $H$.

Cross-entropy is a commonly used loss function in many semantic segmentation tasks and optimizes the model by measuring the difference between the predicted probability distribution of the model and the true distribution. It is also taken as the loss function of MUNet in STSPMNet. The optimization of MUNet aims to minimize the cross-entropy between the ground truth land cover map $X$ and the predicted land cover probabilities by MUNet. The formula for cross-entropy loss is

$$L_{\mathrm{CE}} = -\sum_{c=1}^{C}\sum_{i=1}^{S\times I}\sum_{j=1}^{S\times J} X_{ijc}\cdot\log\left(\hat{P}_{ijc}\right) \tag{10}$$

where $\hat{P}_{ijc}$ is the predicted probability at position $(i, j)$ for land cover type $c$.

Since STSPMNet integrates both super-resolution and semantic segmentation tasks, their loss functions are adopted to simultaneously train and optimize STSPMNet during the training stage. Therefore, the loss function of STSPMNet is designed by combining the loss functions from both SRGAN and MUNet. Consequently, the total loss function of STSPMNet is defined as

$$L = \omega L_{\mathrm{CE}} + \varphi\left(L_{\mathrm{MSE}} + \alpha L_{\mathrm{Per}} + \beta L_{\mathrm{adv}}^{G}\right) \tag{11}$$

where $\alpha$ and $\beta$ represent the hyperparameters that control the weights of each loss, and they are set to empirical values in SRGAN [50], $\omega$ and $\varphi$ denote the coefficients for balancing MUNet and the SRGAN generator, respectively.

### C. Training and Prediction

Both SRGAN and MUNet require end-to-end training data. That is to say, SRGAN uses CR-FR remote sensing image pairs during training, and MUNet requires FR remote sensing images along with their annotated counterparts. For training, a CR remote sensing image at the other date, a FR remote sensing image at the other date, and its classified result as a prior FR land cover map are used as inputs of SRGAN and MUNet. Using the training data as inputs, both SRGAN and MUNet within STSPMNet are incrementally optimized through minimizing the loss function defined in (11). Once SRGAN and MUNet are trained, the CR image at the prediction date is adopted as the input of STSPMNet to predict FR land cover probabilities.

### D. Incorporation of a Prior Land Cover Map

The trained STSPMNet can solely predict FR land cover probabilities from a CR remote sensing image at the prediction date. However, they do not account for changed and unchanged information from bitemporal CR images, which fail to reduce potential uncertainty in the predicted land cover probabilities. Therefore, a temporal constraint extracted from bitemporal CR images by change detection is employed to incorporate a prior FR land cover map at the other date into the FR land cover probabilities, ultimately generating the final FR land cover map at the prediction date.

To analyze land cover changes, two CR fraction images are required for each date. To mitigate the errors introduced by soft classifications, a degradation operation is applied to produce the CR fraction images. The CR fraction images at the other date are derived by degrading the prior FR land cover map, while the CR fraction images at the prediction date are obtained by degrading the predicted FR land cover probabilities. According to bitemporal CR fraction images, three primary land cover change trajectories can be identified as unchanged land cover types, increased land cover types, and decreased land cover types. Inspired by Ling et al. [28], we can assume that for land cover types with unchanged fractions, the FR pixels of these types remain consistent on both dates within each CR pixel. For the land cover change trajectory with an increased fraction of types from the predate to the postdate, it is assumed that the FR pixels of these types at the predate remain unchanged, while the remaining increased FR pixels at the postdate are converted from other land cover types. To incorporate a prior land cover map, a temporal constraint is constructed for the FR pixels with unchanged types in terms of the two land cover change trajectories. Specifically, the land cover type of unchanged FR pixels within a CR pixel remains

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consistent from the prior land cover map to the final land cover map at the prediction date. For the remaining increased and decreased FR pixels within each CR pixel, an optimization approach is developed to determine the optimal land cover type for each FR pixel. According to the maximal spatial dependence principle, this model is constructed to maximize the sum of land cover probabilities of remaining FR pixels, while maintaining the fraction constraints for each land cover type. The optimization model is defined as

$$\text{Max } z = \sum_{c=1}^{C} \sum_{i=1}^{S} \sum_{j=1}^{S} X_{ijc} \cdot \hat{P}_{ijc} \tag{12}$$

$$\text{s.t.} \begin{cases} \sum_{c=1}^{C} X_{ijc} = 1 \\ \sum_{i=1}^{S} \sum_{j=1}^{S} X_{ijc} = N_{\text{rc}} \\ N_{\text{rc}} = S^2 \cdot F_c - N_{\text{uc}} \\ X_{ijc} \in \{0, 1\} \end{cases} \tag{13}$$

where $X_{ijc}$ denotes whether FR pixel $(i, j)$ belongs to land cover type $c$, $F_c$ represents the fraction of a CR pixel associated with land cover type $c$, and $N_{\text{rc}}$ and $N_{\text{uc}}$ are the number of remaining and unchanged FR pixels within a CR pixel for land cover type $c$, respectively. Note that the calculations for both the objective function in (12), and the first two constraint functions in (13) are performed solely on the remaining FR pixels within each CR pixel.

The final FR land cover map at the prediction date can be obtained by combining the temporal constraint with the outcomes of the optimization model for each CR pixel.

### E. Evaluation Metrics

To assess the performance of the proposed STSPMNet, two widely used semantic segmentation evaluation metrics are chosen: overall accuracy (OA) and Kappa coefficient.

## III. Experiments

### A. Experimental Design

Three experiments were conducted to evaluate the performance of the proposed STSPMNet method. The first experiment was performed on a bitemporal Landsat-8 image pair and a prior land cover map derived from a ZY-3 image (https://data.cresda.cn/#/home). The 30-m Landsat-8 images, captured over Jiangyin city, China, were treated as CR images with 940 × 468 pixels, while the 2-m prior land cover map and its corresponding pan-sharpening ZY-3 image with 14 100 × 7020 pixels were used to produce FR images using two testing scale factors of 3 and 5. The 2-m prior land cover map was resampled to generate 10- and 6-m FR land cover maps for scale factors of 3 and 5, respectively. Similarly, the 2-m pan-sharpening ZY-3 image was aggregated into 10- and 6-m FR remote sensing images corresponding to the two scale factors. The second experiment employed a different bitemporal Landsat-8 image pair and a prior land cover map extracted from Sentinel-2 images. The 30-m Landsat-8 images, covering Huangshi city, China, were collected as CR images with 2345 × 2544 pixels, while the 10-m prior land cover map and its corresponding Sentinel-2 image with 7035 × 7632 pixels were used as FR images. The 10-m prior land cover map in the second experiment was generated from Sentinel-2 images by Esri (https://livingatlas.arcgis.com/landcover/). The third experiment also used a bitemporal Landsat-8 image pair and a prior land cover map extracted from GF-1 images. The 30-m Landsat-8 images, covering Tianjin city, China, were collected as CR images with 786 × 927 pixels, while the 2-m prior land cover map and its corresponding GF-1 images with 11 790 × 13 905 pixels were collected to produce FR images using three testing scale factors of 3, 5, and 10. For the scale factors of 3 and 5, FR images were aggregated from 2-m GF-1 images, whereas they were directly generated by resampling for the scale factor of 10. The 2-m prior land cover map over Tianjin city was generated from GF-1 images by the data sharing and service portal (https://data.casearth.cn/dataset/653890f7819aec0f26027c3f). Note that the CR-FR image pairs used in the three experiments were accurately aligned through registration correction during the preprocessing stage.

To train the proposed STSPMNet, we created 1000 training patches for each scale factor in the first experiment. For the scale factor of 3, the CR Landsat-8 image at the other date was randomly cropped into patches with a size of 96 × 96 pixels, while the corresponding areas in the prior FR land cover map and the ZY-3 image were cropped into 288 × 288 pixel patches. For the scale factor of 5, the CR Landsat-8 image was randomly cropped into 64 × 64 pixel patches, and the corresponding patches in the prior FR land cover map and ZY-3 image were resized into 320 × 320 pixels. In the second experiment, we similarly created 5000 training patches for the testing scale factor of 3 and the patches from CR and FR images were cropped in the same manner as in the first experiment for the scale factor of 3. In the third experiment, we similarly created 1000 training patches for the testing scale factors of 3 and 5, and the patches from CR and FR images were cropped in the same manner as in the first experiment for the scale factors of 3 and 5. For the scale factor of 10, the CR Landsat-8 was randomly cropped into 32 × 32 pixel patches, and the corresponding patches in the prior FR land cover map and GF-1 image were resized into 320 × 320 pixels.

During training, the Adam optimizer was employed [7]. The learning rate was set to $3 \times 10^{-4}$ according to the sensitivity analysis in Section IV-A. STSPMNet was updated iteratively for 200 epochs. The coefficients for the loss functions within SRGAN were set to their empirical values $\alpha = 2 \times 10^{-6}$ and $\beta = 10^{-3}$ in (11). Meanwhile, the balancing coefficients $\omega$ and $\varphi$ for MUNet and SRGAN generator were fine-tuned to 0.77 and 0.23, respectively. The proposed STSPMNet was implemented by the PyTorch framework and executed on the RTX 4060 Ti GPU with 16 GB of RAM. To fit within the GPU memory constraints, a batch size of 4 was chosen.

To demonstrate the effectiveness of the proposed STSPMNet, it was compared with a traditional SPM algorithm of subpixel/pixel spatial attraction model (SPSAM) [26], a recent deep learning-based SPM algorithm with convolutional neural networks (SPMCNN) [56], a typical STSPM algorithm of SLCCM [28], and a recent STSRM using local

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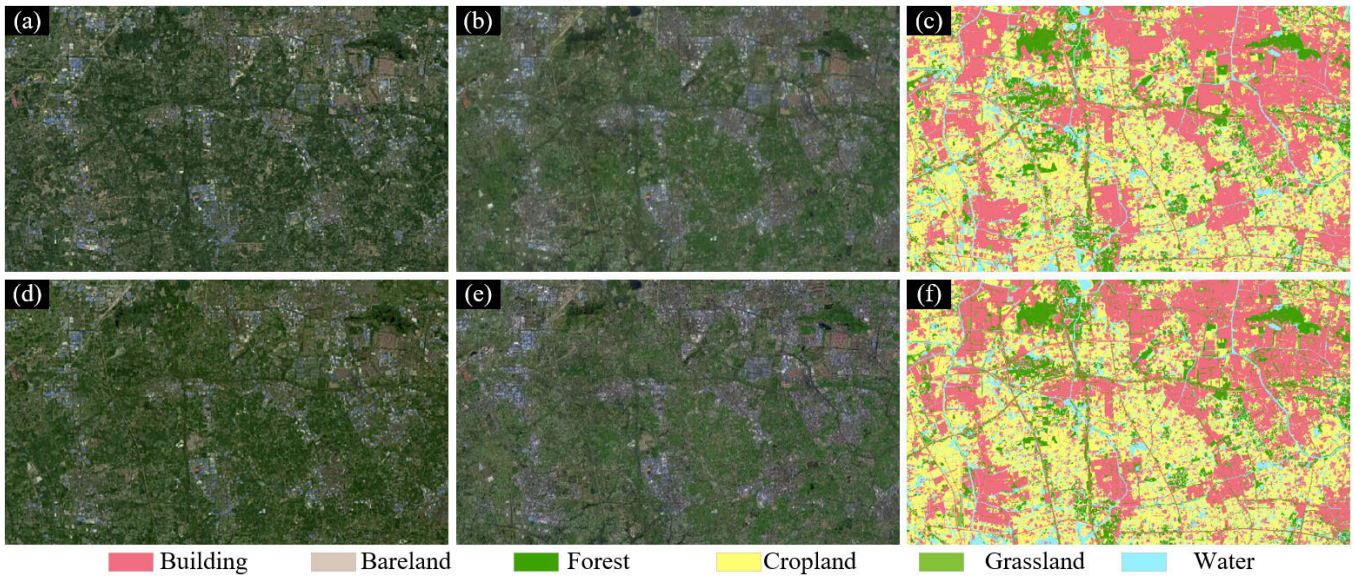



Fig. 3. Dataset for the first experiment. (a) Landsat-8 image acquired in April 2020. (b) ZY-3 image acquired in April 2020. (c) Prior land cover map in 2020. (d) Landsat-8 image acquired in May 2022. (e) ZY-3 image acquired in May 2022. (f) Reference land cover map in 2022.

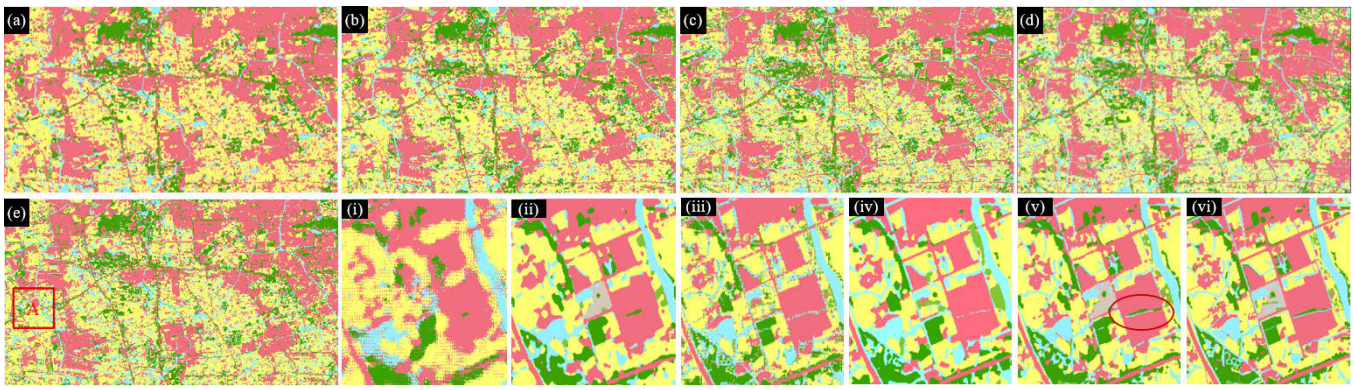



Fig. 4. Results of the first experiment for the scale factor of 3. (a) SPSAM. (b) SPMCNN. (c) SLCCM. (d) LSTSRM. (e) STSPMNet. (i)–(v) Classified results in the subarea of (a)–(e), respectively. (vi) Reference map at the prediction date in the subarea marked by the red rectangle in (e).

spatial-temporal dependence method (LSTSRM) [14]. It is noteworthy that the soft classification results of CR images at the prediction date for both SPSAM and SLCCM methods were obtained using STSPMNet without a super-resolution process from the same training data.

### *B. Experiment 1*

The dataset for the first experiment is shown in Fig. 3. Both the Landsat-8 image in Fig. 3(a) and the pan-sharpening ZY-3 image in Fig. 3(b) were acquired in April 2020 and adopted as the CR-FR image pair at the other date (i.e., before the prediction date) in Jiangyin city, China. The prior land cover map in Fig. 3(c) was manually digitized from the ZY-3 image in Fig. 3(b). The Landsat-8 image at the prediction date (May 2022) was captured in the same region of Jiangyin city, as depicted in Fig. 3(d). The ground truth land cover map in Fig. 3(f) was also obtained through manual digitizing the ZY-3 image at the prediction date in Fig. 3(e) for accuracy assessment.

*1) SPM Results:* The examples of SPM results from the first experiment using a scale factor of 3 are presented in Fig. 4 for illustration and analysis. From the 10-m land cover maps in Fig. 4, it is evident that the five SPM methods yielded a broadly similar spatial distribution of each land cover type in an overall view. However, when focusing on a subarea highlighted by the red rectangle in Fig. 4(e), notable differences in the land cover details of the five SPM results emerge. Both SPSAM and SLCCM generated more speckle noise and jagged boundaries around the land cover patches in this subarea. In contrast, SPMCNN, LSTSRM, and the proposed STSPMNet produced smoother boundaries and less speckle noise. The evident speckle noise in the SPSAM and SLCCM results can be attributed to errors from soft classification, combined with the strict adherence to land cover proportion constraints. SPMCNN suffered from overly smoothed land cover details as it neglected these proportion constraints. SPSAM and SPMCNN, which relied solely on the CR Landsat image at the prediction date, recovered fewer small-sized land cover details compared to SLCCM, LSTSRM, and STSPMNet, which benefited from integrating the prior land cover map. Compared to SLCCM, STSPMNet produced a land cover map with less speckle noise and greater consistency

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TABLE I
ACCURACY OF EACH METHOD IN EXPERIMENT 1

| | | SPSAM | SLCCM | SPMCNN | LSTSRM | STSPMNet |
|---|---|---|---|---|---|---|
| Scale factor=3 | OA (%) | 73.63 | 83.71 | 82.42 | 86.79 | 93.6 |
| | Kappa | 0.6109 | 0.7699 | 0.7436 | 0.8140 | 0.9081 |
| Scale factor=5 | OA (%) | 69.75 | 81 | 82.05 | 85.94 | 90.5 |
| | Kappa | 0.5505 | 0.7326 | 0.7379 | 0.8024 | 0.8638 |

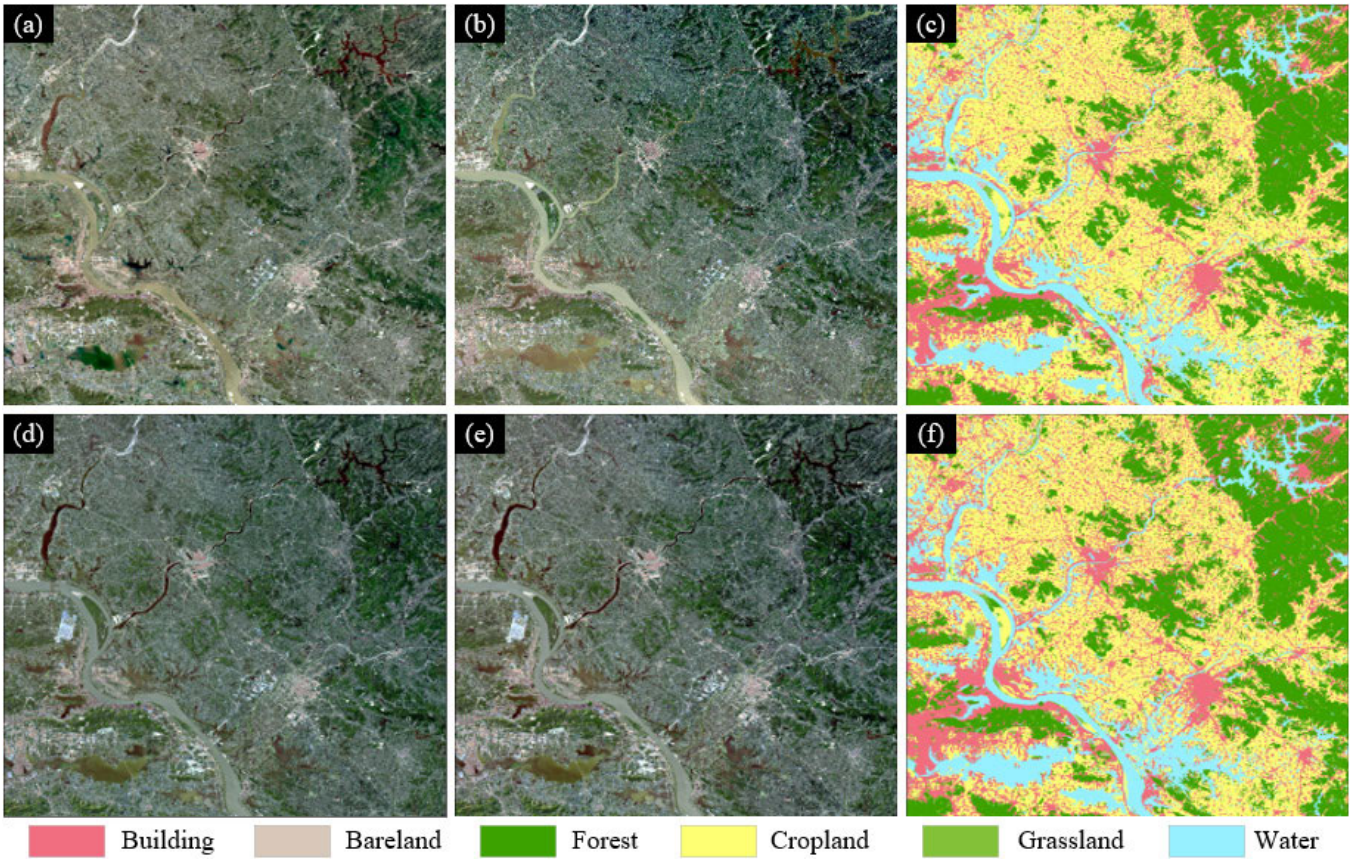



Fig. 5. Dataset for the second experiment. (a) Landsat-8 image acquired in April 2017. (b) Sentinel-2 image acquired in April 2017. (c) Reference land cover map in 2017. (d) Landsat-8 image acquired in March 2020. (e) Sentinel-2 image acquired in March 2020. (f) Prior land cover map in 2020.

with the reference land cover map. Especially, STSPMNet preserved better linear land cover patches (i.e., river and trees) in the area marked by the red circle than SLCCM, as shown in Fig. 4. Compared to LSTSRM, STSPMNet also generated better land cover details, particularly in the area of Fig. 4. Overall, in terms of visual assessment, STSPMNet outperformed the other four SPM methods by generating the best 10-m land cover map from the 30-m CR Landsat image.

*2) Accuracy Assessment:* Two accuracy evaluation metrics for different SPM results were calculated based on the reference land cover maps in Table I for the first experiment. As shown in Table I, the typical SPSAM yielded the lowest OA and Kappa coefficient for the two testing scale factors. SPMCNN and SLCCM exhibited comparable accuracy metrics, with only slight differences in their OA and Kappa coefficients. While LSTSRM achieved higher OA and Kappa coefficient values than SPSAM, SPMCNN, and SLCCM in Table I, it slightly underperformed compared to the proposed STSPMNet in both metrics. The proposed STSPMNet achieved the highest OA and Kappa coefficient among the five methods across both scale factors. Specifically, the OA of STSPMNet was 9.29% and 7.50% higher than the average OA of SPMCNN, SLCCM, and LSTSRM for scale factors of 3 and 5, respectively, with the Kappa coefficient increasing by 0.1323 and 0.1062.

### C. Experiment 2

The dataset for the second experiment is displayed in Fig. 5. The 30-m Landsat-8 image in Fig. 5(a) and 10-m Sentinel-2 image in Fig. 5(b), both acquired in April 2017, are treated as the CR-FR image pair at the prediction date in Huangshi city, China. The 10-m reference land cover map in Fig. 5(c), obtained from the Esri website (https://livingatlas.arcgis.com/landcover/), was employed for accuracy assessment. The Landsat-8 and Sentinel-2 images at the other date of March 2020 (i.e., after the prediction date) were also collected over the same region of Huangshi city, as shown in Fig. 5(d) and (e). The 10-m prior land cover map

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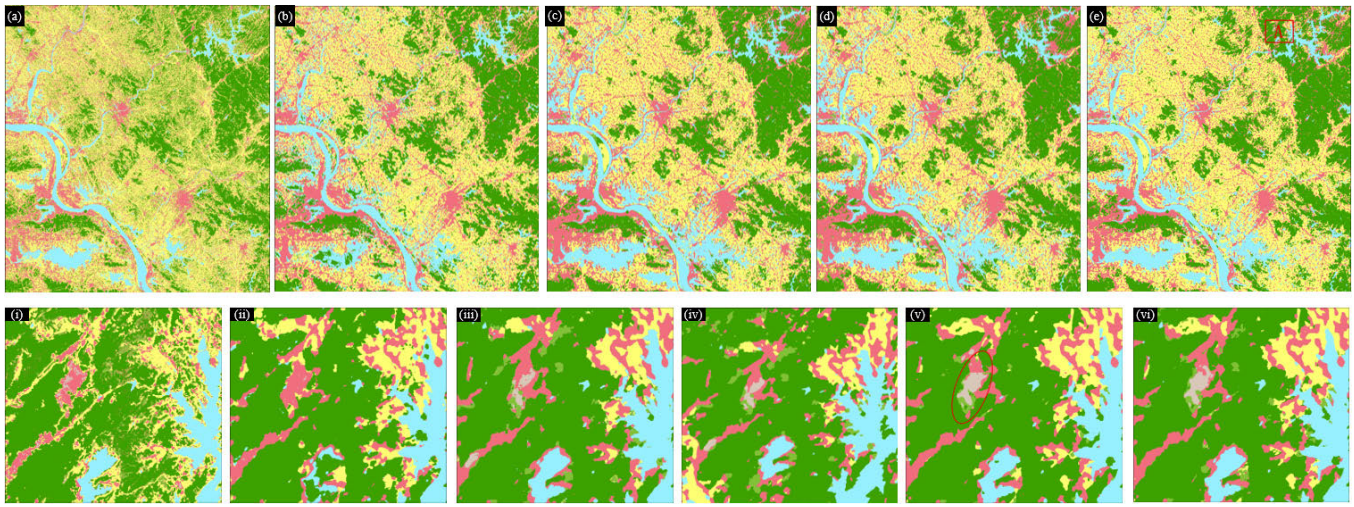


Fig. 6. Results of the second experiment. (a) SPSAM. (b) SPMCNN. (c) SLCCM. (d) LSTSRM. (e) STSPMNet. (i)–(v) Classified results in the subarea of (a)–(e), respectively. (vi) Reference map at the prediction date in the subarea marked by the red rectangle in (e).

in Fig. 5(f) was also similarly sourced from the Esri website and used in the proposed STSPMNet.

*1) SPM Results:* The SPM results of the second experiment using a scale factor of 3 are displayed in Fig. 6. Upon visual examination of the 10-m land cover maps in Fig. 6, the SPSAM result is slightly different from the other SPM results, whereas the other four SPM results appear more consistent overall. However, concentrating on the subarea highlighted by the red rectangle in Fig. 6(e), obvious differences in land cover details emerge across the four methods. Compared to the reference land cover map in Fig. 6(vi), SPSAM generated more speckle noise and misclassified land cover patches in this subarea. In contrast, SPMCNN, SLCCM, LSTSRM, and the proposed STSPMNet produced less speckle noise and land cover maps more closely aligned with the reference map. Despite this, SPMCNN suffers from overly smoothed details and some misclassifications. By integrating a prior land cover map, SLCCM and LSTSRM yielded more accurate results than SPMCNN. Among the methods, STSPM produced the most consistent result with the reference map, particularly for the bareland land cover patches in the red circle of the subarea. Specifically, SPSAM and SPMCNN struggled to restore the bareland land cover patch. In contrast, SLCCM, LSTSRM, and STSPMNet recovered most of it, and STSPMNet restored more accurately than SLCCM and LSTSRM in detail. Therefore, STSPMNet still outperformed the other four SPM methods in generating the most accurate 10-m land cover map from the 30-m CR Landsat image in the second experiment in visual inspection.

*2) Accuracy Assessment:* The accuracy assessment metrics of the five SPM results were obtained from the reference land cover map for the second experiment, as illustrated in Table II. It can be found that the typical SPSAM still yielded the lowest OA and Kappa coefficient, followed by SLCCM, SPMCNN, and LSTSRM. In contrast, the proposed STSPMNet achieved the highest OA and Kappa coefficient. Specifically, the OA of STSPMNet is 11.78% higher than the average OA of the other four methods, with an increase of 0.16305 in the Kappa coefficient.

TABLE II
ACCURACY OF EACH METHOD IN EXPERIMENT 2

| | SPSAM | SLCCM | SPMCNN | LSTSRM | STSPMNet |
|---|---|---|---|---|---|
| OA (%) | 74.34 | 78.93 | 86.06 | 86.28 | 93.18 |
| Kappa | 0.6398 | 0.705 | 0.8084 | 0.8122 | 0.9043 |

### D. Experiment 3

The dataset for the third experiment is displayed in Fig. 7. The Landsat-8 image in Fig. 7(a) and the GF-1 image in Fig. 7(b), both acquired in May 2018, are treated as the CR-FR image pair at the other date (i.e., before the prediction date) in Tianjin city, China. The prior land cover map in Fig. 7(c) was also similarly sourced from the Data Sharing and Service Portal and used in the proposed STSPMNet. The Landsat-8 and GF-1 images at the prediction date of June 2021 were also collected over the same region of Tianjin city, as shown in Fig. 7(d) and (e). The reference land cover map in Fig. 7(f), obtained from the Data Sharing and Service Portal, was employed for accuracy assessment.

*1) SPM Results:* The SPM results of the third experiment using a scale factor of 3 are displayed in Fig. 8. SPSAM and SPMCNN have similar spatial distribution patterns, while SLCCM, LSTSRM, and STSPMNet display greater consistency with each other. When focusing on the subarea marked by the red rectangle in Fig. 8(e), notable differences can be observed in the land cover details among the five SPM results. Compared with the reference land cover map in Fig. 8(vi), SPSAM produced more speckle noise (e.g., buildings) in this subarea than the other four methods. SPMCNN suffered from significant over-smoothing errors (e.g., trees) and failed to reconstruct fine-scale features such as narrow roads and rivers. In contrast, SLCCM, LSTSRM, and the proposed STSPMNet, which leveraged prior land cover information, preserved more land cover details (e.g., rivers and trees) than SPSAM and

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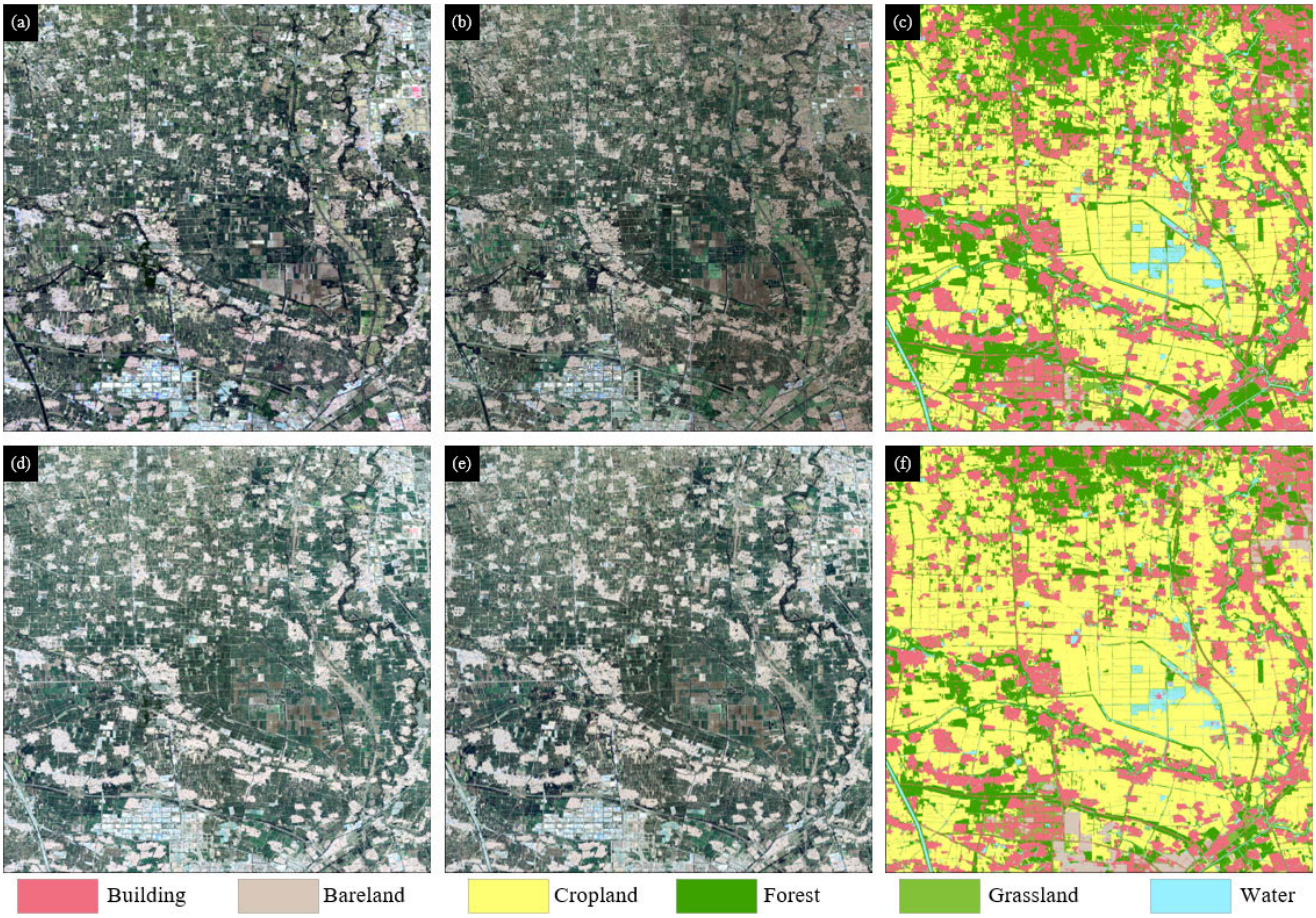



Fig. 7. Dataset for the third experiment. (a) Landsat-8 image acquired in May 2018. (b) GF-1 image acquired in May 2018. (c) Prior land cover map in 2018. (d) Landsat-8 image acquired in June 2021. (e) GF-1 image acquired in June 2021. (f) Reference land cover map in 2021.

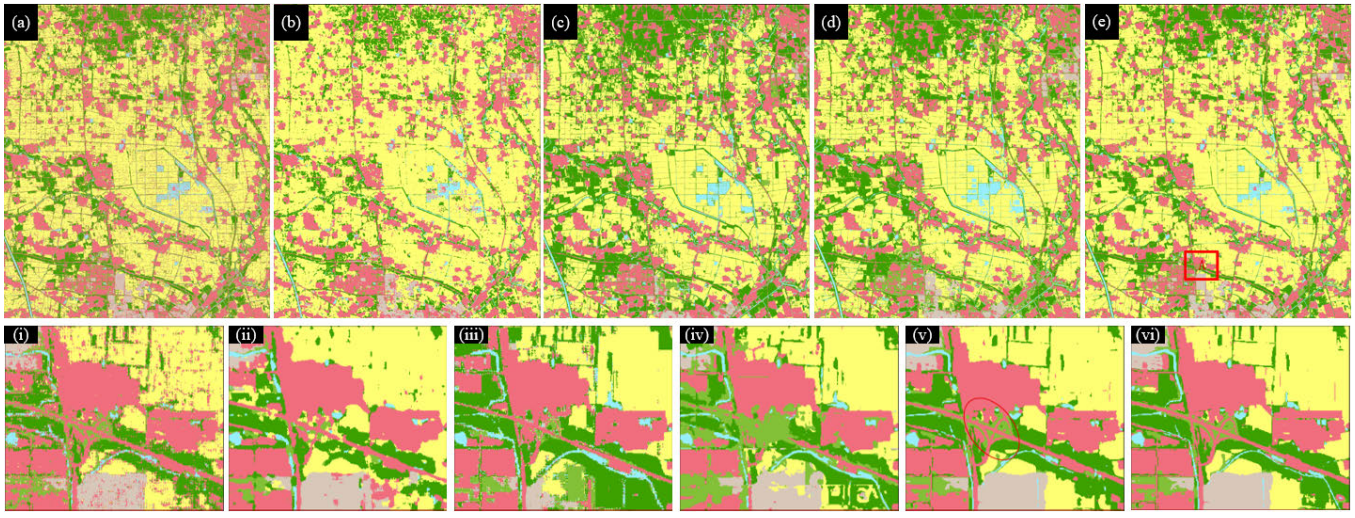



Fig. 8. Results of the third experiment for the scale factor of 3. (a) SPSAM. (b) SPMCNN. (c) SLCCM. (d) LSTSRM. (e) STSPMNet. (i)–(v) Classified results in the subarea of (a)–(e), respectively. (vi) Reference map at the prediction date in the subarea marked by the red rectangle in (e).

SPMCNN. Among them, the proposed STSPMNet still outperformed SLCCM and LSTSRM by more accurately recovering land cover details, such as narrow roads.

*2) Accuracy Assessment:* The accuracy metrics of different SPM methods in the third experiment were calculated based on the reference land cover maps in Table III. It can be found that the typical SPSAM consistently yielded the lowest OA and Kappa coefficients across all three testing scale factors, followed by SLCCM, SPMCNN, and LSTSRM. In contrast, the proposed STSPMNet achieved the highest OA and Kappa values. Specifically, the OA of STSPMNet was 15.81%, 15.39%, and 16.77% higher than the average OA of the other

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TABLE III
ACCURACY OF EACH METHOD IN EXPERIMENT 3

| | | SPSAM | SLCCM | SPMCNN | LSTSRM | STSPMNet |
|---|---|---|---|---|---|---|
| Scale factor=3 | OA (%) | 68.52 | 76.33 | 82.39 | 78.57 | 93.01 |
| | Kappa | 0.5511 | 0.6598 | 0.7242 | 0.6942 | 0.8948 |
| Scale factor=5 | OA (%) | 58.39 | 75.49 | 78.76 | 78.09 | 88.07 |
| | Kappa | 0.4187 | 0.6482 | 0.6772 | 0.6813 | 0.8190 |
| Scale factor=10 | OA (%) | 53.81 | 75.36 | 74.58 | 77.71 | 87.13 |
| | Kappa | 0.3607 | 0.6466 | 0.5965 | 0.6771 | 0.8036 |

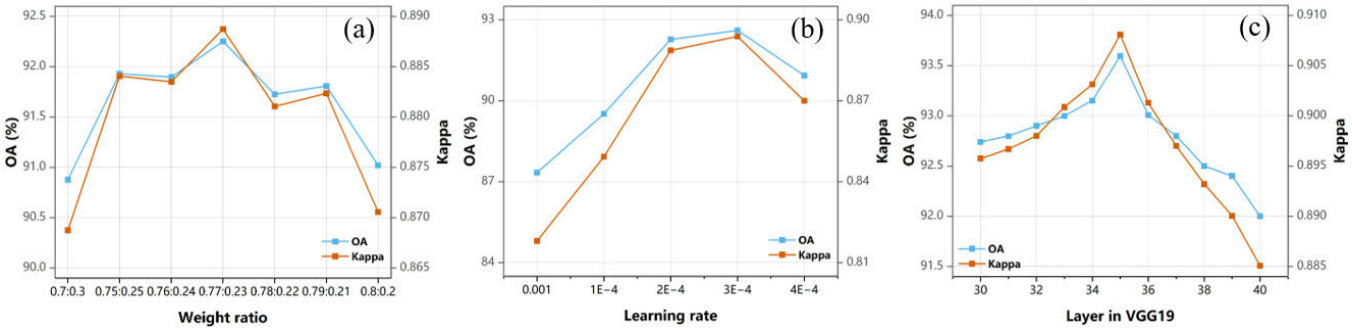



Fig. 9. OA and Kappa curves using different hyperparameters. (a) Weight ratio of loss function. (b) Learning rate. (c) Layer in VGG19 for the perceptual loss function.

four methods, while the Kappa coefficient improved by 0.2375, 0.2127, and 0.2334, respectively.

## IV. DISCUSSION

### A. Sensitivity Analysis for the Hyperparameters

The proposed STSPMNet was optimized by combining the loss functions of SRGAN and MUNet. The parameters $\omega$ for the MUNet loss function and $\varphi$ for the SRGAN loss function were fine-tuned using the first experimental dataset and their sensitivity analysis is illustrated in Fig. 9(a). The OA and Kappa curves indicate that the performance of STSPMNet improves progressively as $\omega$ increases for the MUNet loss function and $\varphi$ decreases for the SRGAN loss function. The OA and Kappa values reach their maximum at $\omega = 0.77$ and $\varphi = 0.23$. Beyond this point, further increases in $\omega$ and decreases in $\varphi$ lead to a gradual decline in the performance of STSPMNet. Therefore, the parameters $\omega$ and $\varphi$ in the total loss function of STSPMNet were set to their optimal values according to fine-tuning.

The learning rate of STSPMNet was also fine-tuned using the first experimental dataset, as shown in Fig. 9(b). The OA and Kappa curves indicate that the performance of STSPMNet improves progressively as the learning rate increases, peaking at 0.0003. Beyond this value, further increases result in a gradual decline in performance; therefore, the optimal learning rate was set to 0.0003 according to Fig. 9(b).

Similarly, the layer index of VGG19 used for the perceptual loss function was fine-tuned, as illustrated in Fig. 9(c). The OA and Kappa values progressively improve with higher layer indices, reaching their maximum at layer 35. Further increases in the layer index lead to a performance decline. Consequently, layer 35 was selected as the optimal setting for the perceptual loss function.

### B. Performance With and Without Prior Land Cover Map

The prior land cover map plays a crucial role as auxiliary data in the proposed STSPMNet, and it is important to investigate its impact on the performance. The performance of STSPMNet, both with and without prior land cover maps, was compared within a subarea of the first experiment. The results of STSPMNet with and without prior land cover maps are illustrated in Fig. 10. Without prior land cover maps, STSPMNet produced smoother boundaries and failed to capture several small-sized land cover patches, especially in the result at the scale of 5. For instance, the land cover patches (i.e., rivers) highlighted by the four black circles in Fig. 8 maintained their linear structure when prior land cover maps were used, but this structure was lost when they were not. The results with prior land cover maps visually align more closely with the reference land cover maps. This difference occurs because most deep learning-based SPM methods tend to produce smooth results, and STSPMNet without prior maps is no exception [47]. Therefore, incorporating prior land cover maps enables STSPMNet to recover more accurate land cover details.

### C. Impact of Scale Factors

The influence of scale factors on the performance of the proposed STSPMNet is significant to analyze its potential for downscaling CR images into FR land cover maps. Based on the results of the first experiment, it was observed that a smaller scale factor (i.e., 3) produced better outcomes compared to a larger scale factor (i.e., 5). Specifically, the smaller scale factor obtained higher OA and Kappa coefficient, as displayed in Table I, and better preserved the detailed structure of land cover patches, as illustrated in Fig. 10. Meanwhile, in the third experiment, both the proposed STSPMNet and the

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TABLE IV
TRAINING AND PREDICTION TIME (S) OF SPMCNN AND STSPMNet IN THE THREE EXPERIMENTS

| | Experiment 1 | | Experiment 2 | | Experiment 3 | |
|---|---|---|---|---|---|---|
| | SPMCNN | STSPMNet | SPMCNN | STSPMNet | SPMCNN | STSPMNet |
| Training | 16 | 191 | 84 | 950 | 16 | 191 |
| Prediction | 10 | 36 | 47 | 253 | 22 | 58 |
| Total | 26 | 227 | 131 | 1203 | 38 | 249 |




Fig. 10. Comparison of STSPMNet results with and without prior land cover maps. The black circled area has better restoration of land cover patches (e.g., rivers) when the proposed STSPMNet is combined with prior land cover maps.

other four methods exhibited lower OA and Kappa coefficients when a larger scale factor was used, as presented in Table III. The performance change of STSPMNet across different scales is generally comparable to other SPM methods. As larger scale factors result in an increase in the number of subpixels, which introduce greater uncertainty and pose challenges in accurately restoring detailed land cover patches. It is also found from Tables I and III that the decrease in accuracy for SPSAM and SPMCNN is significantly greater than that for SLCCM, LSTSRM, and STSPMNet. For example, in the third experiment, the average OA reduction for SPSAM and SPMCNN is 8.69% higher than that of the other three methods when the scale increases from 3 to 10. This performance gap is primarily attributed to the integration of prior land cover maps into SLCCM, LSTSRM, and STSPMNet powered by deep learning and change detection techniques.

### D. Computational Efficiency

The computational efficiency of two deep learning-based SPM methods (SPMCNN and STSPMNet) was evaluated by comparing their training and prediction time. The computational time of the three experiments using the scale factor of 3 is shown in Table IV. It indicates that STSPMNet required nearly ten times longer training time and approximately four times longer prediction time than SPMCNN in all experiments. This difference arises because STSPMNet employed a deeper and more complex convolutional network architecture, which integrated auxiliary CR and FR images. Among the three experiments, the training and prediction time for the first and third experiments are comparable due to their similar image dimensions, while the second experiment needs almost five times longer computational time than the other two experiments.

### E. Impact of Different Land Cover Change Rates

The impact of land cover change rates on the performance of STSPMNet was investigated using three experiments. The change rates are 6.86%, 13.91%, and 21.58% for the first, second, and third experiments, respectively. When comparing the three experimental results, the OA and Kappa values decrease as the land cover change rate increases for SLCCM, LSTSRM, and STSPMNet. For instance, by integrating a prior land cover map for the scale factor of 3, the OA and Kappa values exhibited the most significant decline. These results indicate that higher land cover change rates resulted in greater degradation of the performance of STSPM methods. Notably, STSPMNet demonstrated smaller reductions in OA and Kappa values compared to SLCCM and LSTSRM, despite increasing change rates. Therefore, the integration of deep learning and change detection techniques in STSPMNet reduces the negative impact of high land cover change rates.

### F. Limitations

Although the proposed STSPMNet outperforms four existing SPM methods in reducing uncertainty and reconstructing high-resolution land cover maps, it still has its inherent limitations. First, STSPMNet relies critically on the assumption that there is a perfect spatial alignment for CR-FR image pairs. However, in practice, registration errors may occur due to imperfect correction [58]. Therefore, accurate coregistration of CR-FR image pairs is essential to avoid the negative impact on the performance of STSPMNet. Second, it relies on high-quality prior FR land cover maps to recover fine spatial details; thus, meticulous preparation of these prior data is critical. Third, STSPMNet requires a CR-FR image pair with closely aligned acquisition dates to minimize the negative impacts of land cover changes.

## V. CONCLUSION

This study proposes a new deep STSPMNet that leverages deep learning and change detection techniques to integrate a prior FR land cover map with CR remote sensing imagery to generate FR land cover maps. It combined super-resolution reconstruction and semantic segmentation in a unified framework to realize the end-to-end SPM from CR remote sensing images to FR land cover maps. A change-detection-based optimization approach was developed to integrate the prior FR

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land cover map with the estimated FR land cover probabilities to obtain the final FR land cover map at the prediction date. Both visual and quantitative assessments from three experiments demonstrated that the proposed STSPMNet generated more accurate spatial details of land cover patches and achieved higher accuracy metrics than two SPM methods and two STSPM methods. Furthermore, with the inclusion of a prior FR land cover map, STSPMNet preserved the more accurate structure of land cover patches. Thus, the proposed STSPMNet is a reliable solution for generating accurate FR land cover maps from CR remote sensing images when prior FR land cover maps are available.

## REFERENCES


[1] T. Kasetkasema, M. K. Arora, and P. K. Varshney, "Super-resolution land cover mapping using a Markov random field based approach," *Remote Sens. Environ.*, vol. 96, pp. 302–314, Jun. 2005.

[2] Y. Liu, Y. Zhong, A. Ma, J. Zhao, and L. Zhang, "Cross-resolution national-scale land-cover mapping based on noisy label learning: A case study of China," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 118, Apr. 2023, Art. no. 103265.

[3] Y. Chen, C. Xu, Y. Ge, X. Zhang, and Y. Zhou, "A 100 m gridded population dataset of China's seventh census using ensemble learning and big geospatial data," *Earth Syst. Sci. Data*, vol. 16, no. 8, pp. 3705–3718, Aug. 2024.

[4] K. Karra, C. Kontgis, Z. Statman-Weil, J. C. Mazzariello, M. Mathis, and S. P. Brumby, "Global land use / land cover with Sentinel 2 and deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. IGARSS*, Jul. 2021, pp. 4704–4707.

[5] K. L. Harper et al., "A 29-year time series of annual 300 m resolution plant-functional-type maps for climate models," *Earth Syst. Sci. Data*, vol. 15, no. 3, pp. 1465–1499, 2023.

[6] Y. Chen, J. Zhou, Y. Ge, and J. Dong, "Uncovering the rapid expansion of photovoltaic power plants in China from 2010 to 2022 using satellite data and deep learning," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114100.

[7] Y. Chen et al., "Structure-aware deep learning network for building height estimation," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 137, Mar. 2025, Art. no. 104443.

[8] P. Fisher, "The pixel: A snare and a delusion," *Int. J. Remote Sens.*, vol. 18, no. 3, pp. 679–685, Feb. 1997.

[9] P. M. Atkinson, E. Pardo-Iguzquiza, and M. Chica-Olmo, "Downscaling cokriging for super-resolution mapping of continua in remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 2, pp. 573–580, Feb. 2008.

[10] F. Ling, X. Li, F. Xiao, and Y. Du, "Superresolution land cover mapping using spatial regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4424–4439, Jul. 2014.

[11] P. M. Atkinson, A. Stein, and C. Jeganathan, "Spatial sampling, data models, spatial scale and ontologies: Interpreting spatial statistics and machine learning applied to satellite optical remote sensing," *Spatial Statist.*, vol. 50, Aug. 2022, Art. no. 100646.

[12] Q. Wang and P. M. Atkinson, "The effect of the point spread function on sub-pixel mapping," *Remote Sens. Environ.*, vol. 193, pp. 127–137, May 2017.

[13] Y. Zhong, Y. Wu, X. Xu, and L. Zhang, "An adaptive subpixel mapping method based on MAP model and class determination strategy for hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1411–1426, Mar. 2015.

[14] X. Li et al., "Spatial–temporal super-resolution land cover mapping with a local spatial–temporal dependence model," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4951–4966, Jul. 2019.

[15] S. Wu et al., "Influence of reconstruction scale, spatial resolution and pixel spatial relationships on the sub-pixel mapping accuracy of a double-calculated spatial attraction model," *Remote Sens. Environ., Interdiscipl. J.*, vol. 210, pp. 345–361, Jun. 2018.

[16] Y. Zhang, P. M. Atkinson, X. Li, F. Ling, Q. Wang, and Y. Du, "Learning-based spatial–temporal superresolution mapping of forest cover with MODIS images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 1, pp. 600–614, Jan. 2017.

[17] A. Bateson and B. Curtiss, "A method for manual endmember selection and spectral unmixing," *Remote Sens. Environ.*, vol. 55, no. 3, pp. 229–243, Mar. 1996.

[18] M. Brown, H. G. Lewis, and S. R. Gunn, "Linear spectral mixture models and support vector machines for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 5, pp. 2346–2360, Sep. 2000.

[19] G. M. Foody, "Sharpening fuzzy classification output to refine the representation of sub-pixel land cover distribution," *Int. J. Remote Sens.*, vol. 19, no. 13, pp. 2593–2599, Jan. 1998.

[20] M. Q. Nguyen, P. M. Atkinson, and H. G. Lewis, "Superresolution mapping using a Hopfield neural network with fused images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 3, pp. 736–749, Mar. 2006.

[21] P. M. Atkinson, "Mapping sub-pixel boundaries from remotely sensed images," in *Innovations in GIS*, vol. 4. London, U.K.: Taylor & Francis, 1997, pp. 166–180.

[22] P. Aplin and P. M. Atkinson, "Sub-pixel land cover mapping for per-field classification," *Int. J. Remote Sens.*, vol. 22, no. 14, pp. 2853–2858, Jan. 2001.

[23] Y. Chen, Y. Ge, Y. Chen, Y. Jin, and R. An, "Subpixel land cover mapping using multiscale spatial dependence," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5097–5106, Sep. 2018.

[24] Y. Chen, Y. Ge, G. B. M. Heuvelink, R. An, and Y. Chen, "Object-based superresolution land-cover mapping from remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 328–340, Jan. 2018.

[25] Y. Ge, Y. Chen, A. Stein, S. Li, and J. Hu, "Enhanced subpixel mapping with spatial distribution patterns of geographical objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2356–2370, Apr. 2016.

[26] K. C. Mertens, B. De Baets, L. P. C. Verbeke, and R. R. De Wulf, "A sub-pixel mapping algorithm based on sub-pixel/pixel spatial attraction models," *Int. J. Remote Sens.*, vol. 27, no. 15, pp. 3293–3310, Sep. 2006.

[27] Y. Ge et al., "Principles and methods of scaling geospatial Earth science data," *Earth-Sci. Rev.*, vol. 197, Oct. 2019, Art. no. 102897.

[28] F. Ling, W. Li, Y. Du, and X. Li, "Land cover change mapping at the subpixel scale with different spatial-resolution remotely sensed imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 182–186, Jan. 2011.

[29] X. Li, F. Ling, Y. Du, and Y. Zhang, "Spatially adaptive superresolution land cover mapping with multispectral and panchromatic images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2810–2823, May 2014.

[30] P. Wang, M. Dalla Mura, J. Chanussot, and G. Zhang, "Soft-then-hard super-resolution mapping based on pansharpening technique for remote sensing image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 1, pp. 334–344, Jan. 2019.

[31] M. Q. Nguyen, P. M. Atkinson, and H. G. Lewis, "Superresolution mapping using a Hopfield neural network with LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 2, no. 3, pp. 366–370, Jul. 2005.

[32] F. Ling, Y. Du, F. Xiao, H. Xue, and S. Wu, "Super-resolution land-cover mapping using multiple sub-pixel shifted remotely sensed images," *Int. J. Remote Sens.*, vol. 31, no. 19, pp. 5023–5040, Oct. 2010.

[33] Q. Wang, W. Shi, and L. Wang, "Indicator cokriging-based subpixel land cover mapping with shifted images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 327–339, Jan. 2014.

[34] P. Wang, M. Huang, L. Wang, G. Zhang, H. Leung, and C. Zhao, "Spatiotemporal subpixel mapping based on priori remote sensing image with variation differences," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7556–7575, 2022.

[35] Q. Wang, W. Shi, and P. M. Atkinson, "Spatiotemporal subpixel mapping of time-series images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5397–5411, Sep. 2016.

[36] D. He, Q. Shi, J. Xue, P. M. Atkinson, and X. Liu, "Very fine spatial resolution urban land cover mapping using an explicable sub-pixel mapping network based on learnable spatial correlation," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113884.

[37] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[38] Y. Chen et al., "Edge-enhanced SAM for extracting photovoltaic power plants from remote sensing imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 140, Jun. 2025, Art. no. 104580.

[39] X. Ma, Y. Hong, and Y. Song, "Super resolution land cover mapping of hyperspectral images using the deep image prior-based approach," *Int. J. Remote Sens.*, vol. 41, no. 7, pp. 2818–2834, Apr. 2020.

[40] F. Ling and G. M. Foody, "Super-resolution land cover mapping by deep learning," *Remote Sens. Lett.*, vol. 10, no. 6, pp. 598–606, 2019.


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[41] P. V. Arun, K. M. Buddhiraju, and A. Porwal, "CNN based subpixel mapping for hyperspectral images," *Neurocomputing*, vol. 311, pp. 51–64, Oct. 2018.
[42] X. Zhang et al., "High-quality super-resolution mapping using spatial deep learning," *iScience*, vol. 26, no. 6, Jun. 2023, Art. no. 106875.
[43] Z. Yin, Y. Wu, P. Wu, Z. Hao, and F. Ling, "Super-resolution mapping with a fraction error eliminating CNN model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5408618.
[44] X. Zhang, Y. Ge, F. Ling, J. Chen, Y. Chen, and Y. Jia, "Graph convolutional networks-based super-resolution land cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7667–7681, 2021.
[45] D. He, Q. Shi, X. Liu, and Y. Zhong, "Deep subpixel mapping based on semantic information modulated network for urban land use mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10628–10646, Dec. 2021.
[46] P. Wang, G. Zhang, L. Wang, H. Leung, and H. Bi, "Subpixel land cover mapping based on dual processing paths for hyperspectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1835–1848, Jun. 2019.
[47] D. He, Q. Shi, X. Liu, Y. Zhong, and L. Zhang, "Generating 2m fine-scale urban tree cover product over 34 metropolises in China based on deep context-aware sub-pixel mapping network," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102667.
[48] D. Jia, C. Cheng, C. Song, S. Shen, L. Ning, and T. Zhang, "A hybrid deep learning-based spatiotemporal fusion method for combining satellite images with different resolutions," *Remote Sens.*, vol. 13, no. 4, p. 645, 2021.
[49] Z. Li, Q. Zhu, J. Yang, J. Lv, and Q. Guan, "A cross-domain object-semantic matching framework for imbalanced high spatial resolution imagery water-body extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5629015.
[50] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 105–114.
[51] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.
[52] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[53] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, Jan. 2019, pp. 63–79.
[54] I. Goodfellow et al., "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, 2014, pp. 2672–2680.
[55] Y. Chen, K. Shi, Y. Ge, and Y. Zhou, "Spatiotemporal remote sensing image fusion using multiscale two-stream convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4402112.
[56] D. He, Y. Zhong, X. Wang, and L. Zhang, "Deep convolutional neural network framework for subpixel mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9518–9539, Nov. 2021.
[57] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Represent.*, 2015, pp. 1–14.
[58] S. Hou et al., "RFSDAF: A new spatiotemporal fusion method robust to registration errors," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616018.



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